DESIGN AND ANALYSIS OF ALGORITHMS

FINDING TIME COMPLEXITY

1. Convert the following algorithm into a program and find its time complexity using the counter method.

```
void function (int n)
{
   int i= 1;
   int s =1;
   while(s <= n)
   {
      i++;
      s += i;
   }
}
```

Note: No need of counter increment for declarations and scanf() and count variable printf() statements.

Input:

A positive Integer n

Output:

Print the value of the counter variable

For example:

Input	Result
9	12

ANSWER:

```
#include <stdio.h>
```

```
int count = 0;
void function(int n){
   int i = 1;
   count++;
   int s = 1;
   count++;
   while(s <= n){
      count++;
      i++;
      count++;
      s+=i;
      count++;
   }
   count++;
}
int main(){
   int n;
   scanf("%d", &n);
   function(n);
   printf("%d", count);
   return 0;
}
```

2. Convert the following algorithm into a program and find its time complexity using the counter method.

```
void func(int n)
{
   if(n==1)
   {
```

```
    printf("*");
  }
  else
  {
    for(int i=1; i<=n; i++)
    {
      for(int j=1; j<=n; j++)
      {
        printf("*");
        printf("*");
        break;
      }
    }
  }
}
```

Note: No need of counter increment for declarations and scanf() and count variable printf() statements.

Input:

A positive Integer n

Output:

Print the value of the counter variable

ANSWER:

```
#include <stdio.h>
int count = 0;
void func(int n){
  if(n==1){
    count++;
    //printf("*");
```

```
        count++;
    }
    else{
        count++;
        for(int i = 1; i <= n; i++){
            count++;
            for(int j = 1; j <= n; j++){
                count++;
                //printf("*");
                count++;
                //printf("*");
                count++;
                break;
            }
            count++;
        }
        count++;
    }
}
int main(){
    int n;
    scanf("%d", &n);
    func(n);
    printf("%d", count);
    return 0;
}
```

3. Convert the following algorithm into a program and find its time complexity using counter method.

```
Factor(num) {
  {
    for (i = 1; i <= num;++i)
    {
      if (num % i== 0)
      {
        printf("%d ", i);
      }
    }
  }
```

Note: No need of counter increment for declarations and scanf() and counter variable printf() statement.

Input:

A positive Integer n

Output:

Print the value of the counter variable

ANSWER:

```
#include <stdio.h>
int count = 0;
void Factor(int num){
  for(int i = 1; i <= num; i++){
    count++;
    if(num%i==0){
      //printf("%d", i);
      count++;
    }
    count++;
  }
```

```
    count++;
}
int main(){
    int n;
    scanf("%d", &n);
    Factor(n);
    printf("%d", count);
    return 0;
}
```

4. Convert the following algorithm into a program and find its time complexity using counter method.

```
void function(int n)
{
    int c= 0;
    for(int i=n/2; i<n; i++)
        for(int j=1; j<n; j = 2 * j)
            for(int k=1; k<n; k = k * 2)
                c++;
}
```

Note: No need of counter increment for declarations and scanf() and count variable printf() statements.

Input:

A positive Integer n

Output:

Print the value of the counter variable

ANSWER:

```
#include <stdio.h>
int count = 0;
```

```
void function(int n){
   int c = 0;
   count++;
   for(int i = n/2; i < n; i++){
      count++;
      for(int j = 1; j < n; j = 2 * j){
         count++;
         for(int k = 1; k < n; k = k * 2){
            count++;
            c++;
            count++;
         }
         count++;
      }
      count++;
   }
   count++;
}
int main(){
   int n;
   scanf("%d", &n);
   function(n);
   printf("%d", count);
   return 0;
}
```

5. Convert the following algorithm into a program and find its time complexity using counter method.

```
void reverse(int n)
```

```
{
   int rev = 0, remainder;
   while (n != 0)
   {
      remainder = n % 10;
      rev = rev * 10 + remainder;
      n/= 10;
   }
   print(rev);
}
```

Note: No need of counter increment for declarations and scanf() and count variable printf() statements.

Input:

A positive Integer n

Output:

Print the value of the counter variable

ANSWER:

```
#include <stdio.h>
int count = 0;
void reverse(int n){
   int rev = 0, remainder;
   count++;
   while(n != 0){
      count++;
      remainder = n%10;
      count++;
      rev = rev * 10 + remainder;
      count++;
```

```
        n /= 10;
        count++;
    }
    count++;
    //print(rev);
    count++;
}
int main(){
    int n;
    scanf("%d", &n);
    reverse(n);
    printf("%d", count);
    return 0;
}
```